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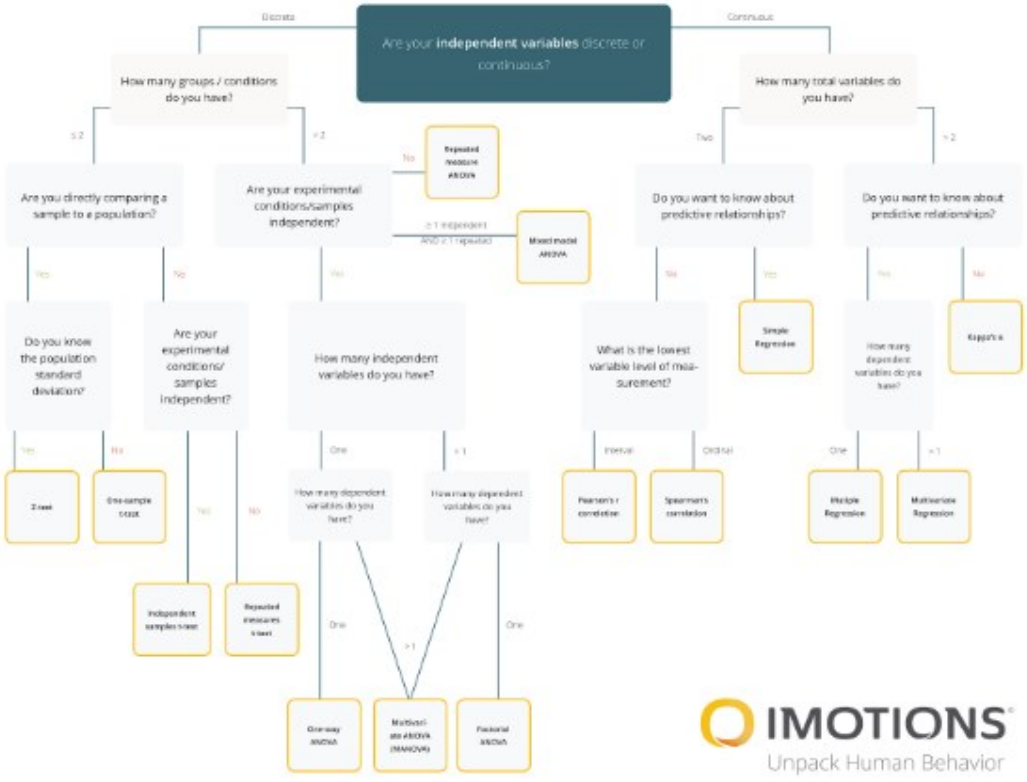
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General Data Analysis and Interpretation

Data Analysis and Interpretation

Statistical Analysis

One of the most challenging aspects of the research process is the data analysis step, which involves determining the appropriate statistical test that is best to evaluate your data. To help with this process, below is a chart that can guide you in selecting the statistical test you need to best determine the results of your experiment and the most appropriate method among common statistical tests:



Descriptive Statistics & Measures of Centrality

After exporting your data from iMotions, choose your preferred software (Excel, SPSS, R, etc,) to conduct basic analysis and start interpreting your data. We recommend that you calculate your descriptive statistics first.

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Descriptive statistics refer to the values that describe your data set generally. For example: What is the average value? How much did the respondents vary in their responses, or did they score mostly similarly? Centrality measures give answers to these questions. The mean, or the average, provides useful information to begin your analyses. You can calculate the means for the different modalities you used in your study, such as the average GSR peaks per minute, the average EEG frontal asymmetry score, the average joy expressed from facial expressions analyses, and so on. Once the averages are laid out, this will provide a starting visualization of how your study groups compare in their responses. What descriptive statistics you will use to represent your data will highly depend on the study that you have run, your business question, and the modalities used. For examples of how we recommend displaying your data for different use cases, download our sample reports here.

Another important value is the *median* of your data set. The median is the value that stands at the midpoint of your data. If you spread the respondents from those who have the lowest GSR peaks to those who have the highest, the median will be the one in the middle. On a perfectly normally distributed sample, the mean and the median are the same. In real research data, this rarely happens. Often, the mean might be biased by the outliers in the data. An outlier is a score that is radically different from the rest of the data set. For instance, if most of your respondents showed a GSR peak per minute response between 2-3, but one of your respondents scored 5 peaks per minute, this will significantly bias the mean you will calculate. So, by either removing the outlier or looking at your median, you will be able to avoid making false interpretations from your data.

Variance

Variance is an important measure to see how differently the respondents scored in a specific measure. Variance value is obtained by subtracting each value from the mean, squaring the differences, and then finding the average of these differences.

Standard Deviation

Mathematically, the standard deviation is the square root of the variance value. Standard deviation reflects how spread out the scores are from their average value in a distribution. A low standard deviation means that the mean is a pretty good reflection of your data since your values do not spread out much. On the other hand, a high standard deviation indicates that the mean may not be a good reflection of your data since the values vary a lot.

Deriving Insights: Inferential Statistical Tests

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Let’s say that after looking at the descriptive statistics, you see that two groups in your study have different means. For instance, you showed two different TV ad videos to two groups and measured their arousal with GSR peaks per minute. You see that the group that saw video 1 has a higher mean GSR peak than the group that saw video 2. How will you decide if the differences are significant enough that you can say video 1 is more arousing?

For this question, inferential statistical tests are used as a further step in the analyses. When interpreting inferential test results, a *p-value* is provided as an output to decide whether group differences are significantly high or not. A p-value is the probability of observing the given results if the difference was due to chance only. In most research, the cut-off point to decide if a p-value shows significance is set to be 0.05, 0.01, or 0.001. If the p-value is below the cutoff, the probability that your results are due to chance is low enough that you can accept them as significant.

t-tests:

t-tests are conducted if the analysis will compare the means of two groups.

Analysis of Variance (ANOVA):

ANOVA’s are conducted if the analysis will compare the means of more than two groups. An ANOVA output provides the mean differences of all groups but only outputs one p-value. If the p-value is lower than the cutoff alpha level, you can infer that at least one group difference is significant. However, the ANOVA itself does not answer which two among your three groups are significantly different. For this further interpretation, a post-hoc test is conducted to see the specific groups that are significantly different.

Common Watch-outs

Multiple comparisons

When conducting statistical analyses, there is always a chance that we find false-positive results due to chance alone. This change is called the type I error, referring to the probability of observing a statistically significant result when there is no significant difference between groups. The more tests we conduct, the higher the chance of observing a false-positive result among our findings. This is why it is recommended to have a clear research question, a hypothesis, and an analysis plan before data collection so that we do as few comparisons as possible.

Reverse inference

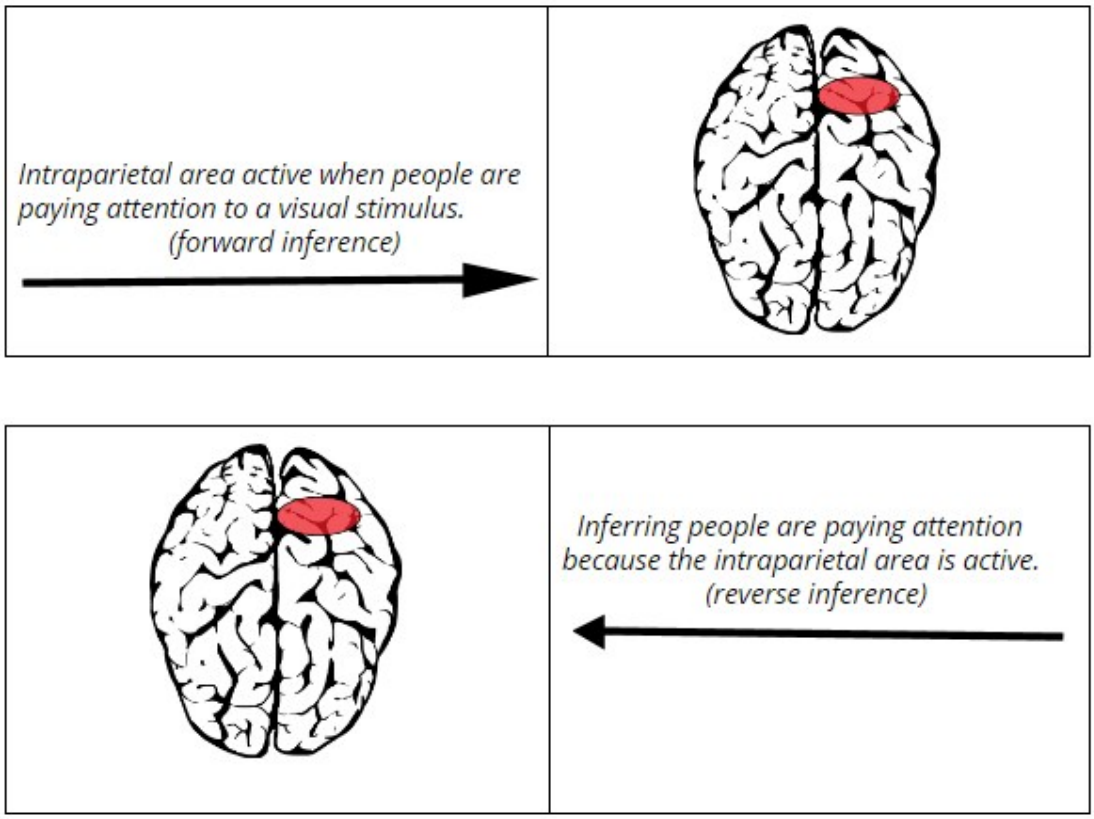
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Reverse inference is a logical fallacy often committed when interpreting brain imaging data. We observe brain regions being active when people are engaged in various cognitive processes. Theoretically, we identify these regions as being "responsible" for a cognitive process. However, the nature of the brain is that many regions are active at the same time and one region is rarely specialized for only one cognitive function. Therefore, we can infer that a brain region associated with a cognitive state must be active when we are engaged in that cognitive process. But, in reverse, if a brain region is activated, we cannot infer the presence of a cognitive process just because that brain region is activated. That would be committing reverse inference.

For example, let us say that brain region X is one of the many regions activated - over and over in multiple different tests and scenarios - when an individual is experiencing the cognitive process of pleasure. From these findings, we will identify brain region X as one of the regions associated with the cognitive processes surrounding pleasure. Now, say we are running a brain imaging experiment, and we observe that brain region X is activated. From this observation, we cannot conclude that the individual under study is experiencing the cognitive process of pleasure. Indeed, this may be the reason that brain region X is activated, but we cannot infer the presence of the specific cognitive process of pleasure simply because brain region X is activated.

Notably, we actually might be correct in inferring a cognitive state from the activation of its associated brain region, but that is a very low chance unless we are dealing with a brain region that is highly exclusive and specialized to one cognitive process. The below image demonstrates forward and reverse inference. The first image shows that we observe that intraparietal regions are reliably active when we engage in visual attention tasks. In the second image, we observe the activation of the same region when engaged in another task. Reverse inference would be to infer that visual attention is engaged in just because this area was observed to be active.

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It is critical to be aware of reverse inference when conducting biometric human behavior research studies. To effectively avoid reverse inference, the first step is to have a scientifically sound and testable research question and hypothesis.

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